

# SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

## ASSESSMENT TOOL 2 – JIGSAW ACTIVITY

|                  |                                                                                                     |               |                                     |
|------------------|-----------------------------------------------------------------------------------------------------|---------------|-------------------------------------|
| Course Code:     | CSA0309                                                                                             | Course Title: | Data Structures                     |
| CO Assessed:     | CO4 – Articulation (BL3)                                                                            | Assessment:   | Assessment Tool 2 – Jigsaw Activity |
| Weightage:       | 35% of CIA                                                                                          | Total Marks:  | 100                                 |
| CO4 Statement:   | Apply hashing strategies for efficient data storage and sorting algorithms for arrangement of data. |               |                                     |
| Student Name:    | R. Rohini Sree                                                                                      | Reg. No.:     | 192524471                           |
| Section / Batch: | 2 <sup>ND</sup> Year AIDS                                                                           | Date:         | 19/08/2024                          |

### Code of Conduct

*I R. Rohini Sree (192524471) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: R. Rohini Sree

### Instructions

- Home Groups (8): 5 students, Expert Groups: One expert per topic. Total: 100 Marks.
- Students first discuss in Expert Groups. They return to Home Groups and teach teammates.
- Each student teaches Concept, Working, Advantages, Disadvantages, Applications, Time Complexity.

| Section / Topic   | Focus                                   | Q Nos     | Marks |
|-------------------|-----------------------------------------|-----------|-------|
| Hashing & Sorting | Hashing, Resolution Techniques, Sorting | 8 Teams   | 100   |
| GRAND TOTAL:      |                                         | 100 Marks |       |

## Annexure A – Jigsaw Activity

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1. Scenario 1 – Airport Baggage System (Every luggage has Barcode)
  - a. Hashing Method
  - b. Collision handling
  - c. Fast luggage
2. Scenario 2 – E-Commerce Warehouse (Products stored by SKU)
  - a. Hashing strategy
  - b. Inventory retrieval
  - c. Sorting inventory.
3. Scenario 3 – Traffic Police Database (Vehicle Registration Numbers)
  - a. Fast searching
  - b. Duplicate handling
  - c. Sorting violations
4. Scenario 4 – Social Media Platform (Millions of users)
  - a. Hash Function
  - b. Collision handling – Separate Chaining
  - c. Fast searching
5. Scenario 19 – University Admission Portal (A university receives 75,000 applications. Applicants are searched using Application IDs, while merit lists are prepared based on entrance scores.)
6. Scenario 22 – Pharmacy Inventory Management (A pharmacy maintains thousands of medicine records. Pharmacists search medicines by code, while stock reports are arranged based on expiry dates.)
7. Scenario 24 – Weather Monitoring System (A weather station collects data from thousands of sensors every minute. Scientists retrieve records using sensor IDs and generate reports sorted by temperature and rainfall.)
8. Scenario 6 – Digital Music Streaming App (Songs, Artist, Playlist, Genre)
  - a. Fast searching
  - b. Hash Function
  - c. Fast Sorting

## Rubrics for Calculation

| Criteria                                        | Weightage | Level 4 – Exemplary                                                                         | Level 3 – Proficient                                       | Level 2 – Developing                                        | Level 1 – Beginning                                           |
|-------------------------------------------------|-----------|---------------------------------------------------------------------------------------------|------------------------------------------------------------|-------------------------------------------------------------|---------------------------------------------------------------|
| <b>Problem Understanding</b>                    | 20        | Accurately interprets all problem requirements and constraints before coding.               | Understands most requirements with minor omissions.        | Partially understands the problem; misses some constraints. | Misinterprets the problem or ignores key requirements.        |
| <b>Correctness of Solution</b>                  | 30        | Program compiles successfully and passes all hidden and visible test cases.                 | Program passes most test cases with minor errors.          | Program passes some test cases but has logical issues.      | Program fails to compile or produces incorrect results.       |
| <b>Data Structure &amp; Algorithm Selection</b> | 20        | Selects the most appropriate data structure and algorithm with an optimized solution.       | Chooses an appropriate solution with minor inefficiencies. | Uses a workable but less efficient approach.                | Uses an inappropriate data structure or algorithm.            |
| <b>Code Quality &amp; Documentation</b>         | 20        | Well-structured, modular, readable, and properly indented code with meaningful identifiers. | Readable code with minor formatting issues.                | Basic coding style with inconsistent formatting.            | Poorly organized code that is difficult to understand.        |
| <b>Time &amp; Space Optimization</b>            | 10        | Demonstrates excellent optimization and correctly analyses time and space complexity.       | Good optimization with minor inefficiencies.               | Limited optimization and incomplete complexity analysis.    | No optimization and incorrect or missing complexity analysis. |

| Faculty In-charge | Course Coordinator | Program Director |
|-------------------|--------------------|------------------|
| Signature: _____  | Signature: _____   | Signature: _____ |
| Date: _____       | Date: _____        | Date: _____      |

